Check for updates

Econometrica, Vol. 87, No. 3 (May, 2019), 699–739

A DISTRIBUTIONAL FRAMEWORK FOR MATCHED EMPLOYER EMPLOYEE DATA

STÉPHANE BONHOMME
Department of Economics, University of Chicago

THIBAUT LAMADON
Department of Economics, University of Chicago

ELENA MANRESA
Department of Economics, New York University

We propose a framework to identify and estimate earnings distributions and worker composition on matched panel data, allowing for two-sided worker-firm unobserved heterogeneity and complementarities in earnings. We introduce two models: a static model that allows for nonlinear interactions between workers and firms, and a dynamic model that allows, in addition, for Markovian earnings dynamics and endogenous mobility. We show that this framework nests a number of structural models of wages and worker mobility. We establish identification in short panels, and develop tractable two-step estimators where firms are classified in a first step. Applying our method to Swedish administrative data, we find that log-earnings are approximately additive in worker and firm heterogeneity. Our estimates imply the presence of strong sorting patterns between workers and firms, and a small contribution of firms—net of worker composition—to earnings dispersion. In addition, we document that wages have a direct effect on mobility, and that, beyond their dependence on the current firm, earnings after a job move also depend on the previous employer.

KEYWORDS: Two-sided heterogeneity, bipartite networks, matched employer employee data, sorting, job mobility.

1. INTRODUCTION

IDENTIFYING THE CONTRIBUTIONS of worker and firm heterogeneity to earnings dispersion is an important step toward answering a number of economic questions, such as the nature of sorting patterns between heterogeneous workers and firms or the sources of earnings inequality.

Two influential literatures have approached these questions from different angles. The method of Abowd, Kramarz, and Margolis (1999) (AKM hereafter) relies on two-way fixed-effects regressions to account for unobservable worker and firm effects, and allows one to quantify their respective contributions to earnings dispersion, as well as correlations between worker and firm effects. The AKM method is widely used in labor and other fields in economics.[1] A second literature tackles similar issues from a structural perspec-

Stéphane Bonhomme: sbonhomme@uchicago.edu
Thibaut Lamadon: lamadon@uchicago.edu
Elena Manresa: em1849@nyu.edu

We thank the four anonymous referees, our discussants Pat Kline, Chris Hansen, and Mikkel Sølvsten, and Joe Altonji, Bryan Graham, Jeremy Lise, Rafael Lopes de Melo, Costas Meghir, Magne Mogstad, Derek Neal, Luigi Pistaferri, Jean-Marc Robin, Adam Rosen, Raffaele Saggio, Uta Schoenberg, Robert Shimer, Chris Taber, and audiences at various places for useful comments. We thank the IFAU for access to, and Lisa Laun for help with, the Swedish administrative data. The authors acknowledge support from NSF Grant SES-1658920.

[1]Applications of the method to earnings data include Gruetter and Lalive (2009), Mendes, van den Berg, and Lindeboom (2010), Woodcock (2008), Card, Heining, and Kline (2013), Goldschmidt and Schmieder

© 2019 The Econometric Society https://doi.org/10.3982/ECTA15722

tive, by developing and estimating fully specified theoretical models of sorting in the labor market.[2]

Reconciling these reduced-form and structural literatures has proven difficult. While the AKM method provides a tractable way to deal with two-sided unobserved heterogeneity, the AKM model relies on substantive, possibly restrictive assumptions. The absence of interactions between worker and firm attributes restricts complementarity patterns in earnings. This is at odds with the theoretical literature which, since Becker (1973), has emphasized the link between complementarity and sorting (Shimer and Smith (2000), Eeckhout and Kircher (2011)). In addition, the AKM model is static, in the sense that worker mobility does not depend on earnings realizations conditional on worker and firm heterogeneity, and that earnings after a job move do not depend on the previous firm. Such a static model is not able to account for a number of mechanisms that have been emphasized in the dynamic structural literature.

On the other hand, attempts at structurally estimating dynamic models of sorting have faced computational and empirical challenges. The dimensions involved are daunting: how to estimate a model of worker mobility and earnings with hundreds of thousands of workers and dozens of thousands of firms in the presence of both firm and worker unobserved heterogeneity? And how much of the results are driven by functional form assumptions?

In this paper, we introduce an empirical framework with two-sided unobserved heterogeneity that nests a range of theoretical mechanisms emphasized in the literature. While allowing for rich patterns of complementarities, sorting, and dynamics, the framework preserves parsimony using a dimension reduction technique to model firm heterogeneity. We propose two models, static and dynamic, which allow for interaction effects between worker and firm heterogeneity. In the dynamic model, we let job mobility depend on earnings realizations in addition to worker and firm attributes, and we allow earnings after a job move to depend on attributes of the previous firm beyond those of the current one by specifying dynamic persistence as first-order Markov.

We provide conditions for identification in short panels under discrete worker heterogeneity. The primary source of identification is given by job movers. For the static model we rely on two periods, while we use four periods to identify the dynamic model. The ability of our method to deal with short panels is important, since even when matched employer employee data sets with a long panel dimension are available, assuming time-invariant heterogeneity of either workers or firms over long periods may be unattractive. Our analysis shows that mobility and heterogeneity patterns play a key role in identifying complementarities.

We define the relevant level of firm unobserved heterogeneity as the *class* of a firm. We model worker types as draws from a discrete distribution, and allow for unrestricted interactions between worker types and firm classes. In principle, a class could be a firm itself. However, in typical matched employer employee data sets, the number of job movers

(2015), Song, Price, Guvenen, and Bloom (2015), and Sorkin (2018), among others. The AKM estimator has been used in a variety of other fields, for example to link banks to firms, teachers to schools or students, and to document differences across areas in patients' health care utilization (e.g., Kramarz, Machin, and Ouazad (2015), Jackson (2013), Finkelstein, Gentzkow, and Williams (2016)).

[2]Many structural models proposed in the literature build on Becker (1973). Examples are Lopes de Melo (2018), Lise, Meghir, and Robin (2016), Bagger and Lentz (2014), Hagedorn, Law, and Manovskii (2017), and Lamadon, Lise, Meghir, and Robin (2013).

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

per firm tends to be small, which creates an incidental parameter bias in estimation.[3] In such environments, reducing the number of classes can alleviate small-sample biases. We use a k-means clustering estimator to classify firms based on how similar their earnings distributions are. The classification may also be based on mobility patterns or longitudinal earnings information, and it can be modified to incorporate firm characteristics such as value added. We establish the consistency of the classification under discrete firm heterogeneity.[4]

We use a two-step approach for estimation. In the *classification* step, we group firms into classes using k-means clustering, and in the *estimation* step, we estimate the model by maximum likelihood, conditional on the estimated firm classes. Estimating firm classes in a first step is helpful for tractability. We verify in simulations that our estimator performs well in data sets similar to the one of our application. We also confirm the ability of our estimator to recover wage functions in data sets generated according to the theoretical model of Shimer and Smith (2000), extended to allow for on-the-job search, under both positive and negative assortative matching.

We take our approach to Swedish matched employer employee panel data, focusing on males for the 2002–2004 period. The estimates of our static model imply that an additive specification provides a good first-order approximation to log-earnings, although our results also highlight the presence of some complementarities between firms and lower-type workers. Between-firm differences explain 38% of the overall log-earnings variance. However, our estimates imply that, net of the effect of worker composition, firm heterogeneity accounts for less than 5% of the overall variance (i.e., less than 13% of the between-firm variance). The largest share of the variance is explained by worker heterogeneity. In addition, we find a strong association between worker and firm heterogeneity, with a correlation ranging between 30% and 50% depending on the specification.

These results suggest that similar workers are not paid very differently across employers, although different workers tend to work in very different firms. The presence of strong sorting, together with the absence of strong complementarities in wages, is difficult to reconcile with models where sorting is driven by complementaries in production, as in Becker (1973). Alternative explanations for sorting have been proposed, such as the presence of amenities, peer effects, or more complex heterogeneity, although our findings might also be partly driven by specificities of the Swedish labor market.

The estimates of our dynamic model in the 2001–2005 period, besides being in line with the cross-sectional variance decomposition implied by our static model, shed light on several mechanisms that have been emphasized in the structural literature. In particular, we find that low earnings realizations, conditional on worker and firm heterogeneity, tend to make workers more likely to move. This violation of exogenous mobility may indicate the presence of match heterogeneity. We also find evidence of an effect of the previous employer on current earnings, conditional on the current firm's class. This state depen-

[3]See Abowd, Kramarz, Lengermann, and Pérez-Duarte (2004), Andrews, Gill, Schank, and Upward (2008, 2012), and recently Kline, Saggio, and Sølvsten (2018) for methods to address incidental parameter bias in fixed-effects regressions.

[4]Similarly as in most of the literature on discrete estimation, this result is derived under the assumption that the population of firms consists of a finite, known number of classes. In Bonhomme, Lamadon, and Manresa (2017), we considered a setting where the discrete modeling is viewed as an approximation to an underlying, possibly continuous, distribution of firm unobserved heterogeneity, and we provided consistency results and rates of convergence.

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

dence effect could be rationalized by existing theories, such as the offer and counteroffer mechanism of Postel-Vinay and Robin (2002).

Literature and Outline

The methods we propose contribute to a large literature on the identification and estimation of models with latent heterogeneity. Discrete fixed-effects approaches have recently been proposed in single-agent panel data analysis (Hahn and Moon (2010), Lin and Ng (2012), Bonhomme and Manresa (2015)). The k-means clustering algorithm we use to classify firms is widely used in a number of fields, and efficient computational routines are available (Steinley (2006)). Here, we use such an approach in models with two-sided heterogeneity. Our approach to identification and estimation of mixture models has a number of precedents in the literature, such as Hall and Zhou (2003), Hu (2008), Henry, Kitamura, and Salanié (2014), Levine, Hunter, and Chauveau (2011), Bonhomme, Jochmans, and Robin (2014), and Hu and Schennach (2008) and Hu and Shum (2012) for continuous mixtures. Our conditional mixture approach is also related to mixed membership models (Blei, Ng, and Jordan (2003), Airoldi, Blei, Fienberg, and Xing (2008)).

Compared to this previous work, we rely on a hybrid "one-sided correlated random-effects" approach, where we model the firm classes as discrete fixed effects, and the worker types as (discrete or continuous) random effects correlated with the firm classes. This approach is motivated by the structure of typical matched employer employee data sets. With sufficiently many workers per firm, firm class membership will be accurately estimated. In contrast, the number of observations for a given worker is typically small. This approach can alleviate the incidental parameter bias of fixed-effects estimators, particularly in short panels. It also offers a tractable way of allowing for complementarities and dynamics. Bonhomme (2017) reviewed existing econometric methods for bipartite network data.

Also related, Abowd, McKinney, and Schmutte (2018) proposed a Bayesian approach where both firm and worker heterogeneity are discrete. Their setup allows for latent match effects to drive job mobility, in a way that is related to—but different from—our dynamic model. Hagedorn, Law, and Manovskii (2017) proposed to recover worker types by ranking workers by their earnings within firms, and aggregating those partial rankings across firms. Their method relies on long panels, and exploits the implications of a structural model to identify firm heterogeneity. In contrast, while our framework nests a number of theoretical models of wages and mobility, it is not tied to a specific structural model.

The outline of the paper is as follows. In Section 2, we present the framework. In Sections 3 and 4, we study identification and estimation. In Sections 5 and 6, we show empirical results based on the static and dynamic models. Last, we conclude in Section 7. The Supplemental Material (Bonhomme, Lamadon, and Manresa (2019)) contains details on computation and several extensions, and an exercise on simulated data generated using a theoretical sorting model.

2. FRAMEWORK OF ANALYSIS

We consider an economy composed of N workers and J firms. We denote as j_{it} the identifier of the firm where worker i is employed at time t. Job mobility between a firm at t and another firm at $t+1$ is denoted as $m_{it} = 1$.

Heterogeneity across firms is characterized by their *class*. We denote as $k_{it} = k(j_{it})$ in $\{1, \dots, K\}$ the class of firm j_{it}. The K classes form a partition of the set of firms. There may be as many classes as firms, in which case $K = J$ and $k_{it} = j_{it}$. Alternatively, firm classes could be defined in terms of observables such as industry or size. In Section 4, we

14680262, 2019, 3, Downloaded from https://onlinelibrary.wiley.com/doi/10.3982/ECTA15722 by Tel Aviv University, Wiley Online Library on [09/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License